Seg Object 1

2D BBOX

Blender Python

```
import bpy
import math

# Build model
bpy.ops.mesh.primitive_add(
    vertices=vertices,
    radius=radius,
    location=location,
    rotation=(0, 0, 0)
)
...

# Render and save
bpy.ops.render.render()
```

3D Modeling

Selected

3D Candidates

The keyboard buttons extend beyond the black keyboard area.

Refine

Blender Python

```
import bpy


FPS = 30
scene.render.ffmpeg.format = 'MPEG4'

# Build Articulation
glass_lid["joint_type"] = "prismatic"
glass_lid["joint_parent"] = "pot"
...

# Render animation
bpy.ops.render.render(animation=True)
```

Articulation Animation

Scene Image

3D BBOX

Object URDF

```
{
"scene_tree": {
    "Desk": {
        "children": ["Laptop"]
    }
},
"objects_data": [
    {
        "name"
        "description"
        "is_environment_support"
        "is_occluded"
        "2d_bbox"
        "3d_bbox"
    }
]}
```

Object-Level LOOP

Pressable

Refined Articulation Demostration Video

Wrong Axis: The screen axis position is incorrect.
Miss Joint: The keyboard keys should also be pressable.

Refine

4 error types:

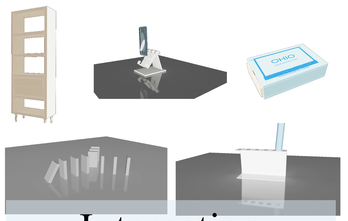
redundent joint/miss joint

wrong axis/wrong range

Articulation Demostration Video

Issac Sim

Scene-Level LOOP

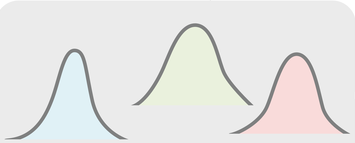


Interaction

Final Scene

```
{
"scene_tree": {
    "Desk": {
        "children": ["Laptop"]
    }
},
"objects_data": [
    {
        "name"
        "description"
        "is_environment_support"
        "is_occluded"
        "2d_bbox"
        "3d_bbox"
    }
]}
```

Scene Layout Tree



Initial Scene

Learning Distribution Using Simulation-based Reward

Cross Entropy Method

Simulator Closed-Loop